

Modeling temperature-dependent life-cycle toxicity of thiamethoxam in *Chironomus riparius* using a DEB-TKTD model

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ABSTRACT

The neonicotinoid insecticide thiamethoxam (TMX) is widely used to protect crops against insect pests. Despite some desirable properties such as its low toxicity to birds and mammals, concerns have been raised about its toxicity to non-target arthropods, including freshwater insects like chironomids. Whereas multiple studies have investigated chronic effects of neonicotinoids in chironomid larvae at standardized laboratory conditions, a better understanding of their chronic toxicity under variable temperatures and exposure is needed for coherent extrapolation from the laboratory to the field. Here, we developed a quantitative mechanistic effect model for *Chironomus riparius*, to simulate the species' life history under dynamic temperatures and exposure concentrations of TMX. Laboratory experiments at four different temperatures (12, 15, 20, 23 °C) and TMX concentrations between 4 and 51 µg/L were used to calibrate the model. Observed concentration-dependent effects of TMX in *C. riparius* included slower growth, later emergence, and higher mortality rates with increasing concentrations. Furthermore, besides a typical accelerating effect on the organisms' growth and development, higher temperatures further increased the effects associated with TMX. With some data-informed modeling decisions, most prominently the inclusion of a size dependence that makes larger animals more sensitive to TMX, the model was parametrized to convincingly reproduce the data. Experiments at both a constant (20 °C) and a dynamically increasing temperature (15–23 °C) with pulsed exposure were used to validate the model. Finally, the model was used to simulate realistic exposure conditions using two reference exposure scenarios measured in Missouri and Nebraska, utilizing a moving time window (MTW) and either a constant temperature (20 °C) or the measured temperature profiles belonging to each respective scenario. Minimum exposure multiplication factors leading to a 10 % effect (EP₁₀) in the survival at pupation, i.e., the most sensitive endpoint found in this study, were 25.67 and 21.87 for the Missouri scenario and 38.58 and 44.64 for the Nebraska scenario, when using the respective temperature assumptions. While the results illustrate that the use of real temperature scenarios does not systematically modify the EP_x in the same direction (making it either more or less conservative when used as a risk indicator), the advantage of this approach is that it increases the realism and thus reduces the uncertainty associated with the model predictions.

1. Introduction

Thiamethoxam (TMX) is a systemic insecticide from the neonicotinoid class that has foliar, soil and seed treatment uses to protect a wide range of crops against insect pests. Like other neonicotinoids, TMX acts on the target insects by irreversibly binding to nicotinic acetylcholine receptors (nAChRs) in their central nervous system, causing

nervous overstimulation that leads to paralysis and, ultimately, the death of the organism (Goulson, 2013). A desired property of neonicotinoids that sets them apart from other insecticide classes is their relatively low toxicity to birds and mammals. However, concerns have been raised about their acute and chronic toxicity to non-target arthropods, including freshwater species that may be exposed to insecticides via runoff to surface waters. If freshwater arthropod species

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are likely to be directly adversely affected, other species may be indirectly affected. In the USA, prior to its registration review under the Federal Insecticide, Fungicide, and Rodenticide Act (FIFRA), direct and indirect effects of a plant protection product must be determined for all species federally listed as threatened or endangered under the Endangered Species Act (ESA). In a study by Raby et al. (2018a), comparing the acute toxicity of neonicotinoids toward a wide range of aquatic arthropods, diptera were identified among the most sensitive taxa, which was later also confirmed for chronic toxicity (Raby et al., 2018b). Studies investigating chronic sublethal effects of neonicotinoids in larvae of *Chironomus* spp. reported, among others, a decrease in growth, a delay in the time until emergence, and a lower rate of successful emergence of the midges when compared to the control treatments (Saraiva et al., 2017; Cavallaro et al., 2017; Langer-Jaesrich et al., 2010). Whereas these studies were all performed under constant laboratory conditions, exposure to TMX occurs under environmentally dynamic conditions as it can be applied from late spring to mid-summer across a range of geographies. Therefore, a better understanding of chronic neonicotinoid toxicity under realistic field conditions would greatly benefit the environmental risk assessment for these chemicals. Especially, a better understanding of the role of temperature in the organisms' life history and their sensitivity to a neonicotinoid is crucial and needs to be considered if laboratory results are to be linked to the field in a realistic way. Previous research has shown that temperature is an important factor influencing the acute and chronic toxicity of insecticides in various species, with higher temperatures typically associated with increased toxicity (e.g., Abbes et al., 2015; Huang et al., 2023; Muturi et al., 2011). However, even with the right data, laboratory-to-field extrapolation is not straightforward, particularly when dynamic conditions are taken into account.

Mechanistic effect models, describing the complete life history of an organism in dependence of one or multiple stressors, are broadly recommended as tools for extrapolating results from static laboratory setups to untested dynamic conditions in the field (EFSA, 2018; Sherborne et al., 2020; Galic et al., 2010). Dynamic Energy Budget (DEB) models in particular have received recognition due to their rigorous adherence to physical principles (e.g., the law of conservation of energy) and their high genericity, allowing them to be parametrized to virtually any species of interest (Kooijman, 2009). Furthermore, DEB models can be extended by toxicokinetic-toxicodynamic (TKTD) submodels, capturing the interactions between an organism and a chemical stressor that ultimately lead to sublethal and lethal effects, reflected in deviations of model state variables from the control (Jager et al., 2023). As a bonus, a simple standard framework exists within the DEB theory that describes how its physiological rates can be related to environmental temperatures using the Arrhenius equation. However, the link between temperature and individual TKTD processes is less clear and requires further investigation (Heugens et al., 2003; Mangold-Döring et al., 2022; Rakel et al., 2022).

In this study, we aimed to evaluate effects to *Chironomus riparius* from realistic exposure in the field by accounting for realistic thermal conditions. To this end, we developed a mechanistic model for the temperature-dependent effects of TMX in *C. riparius*, which was calibrated and validated with laboratory experiments at both constant and dynamic conditions of temperature and TMX exposure. The model design combines a physiological (DEB) model with TKTD extensions, including process-specific temperature dependencies. Simulations at realistic exposure conditions, both in terms of TMX concentrations and water temperatures, were performed using a moving time window approach (EFSA, 2018; Sherborne et al., 2020) for reference scenarios measured in streams in Missouri and Nebraska. This approach allowed us to determine the most adverse exposure window per scenario and exposure multiplication factors (EP_x), leading to relevant effect levels.

2. Materials and methods

2.1. Chronic toxicity tests with *C. riparius*

Chronic toxicity tests, based on OECD test guidelines 219 and 233 (OECD, 2023, 2010), were used to investigate the effects of TMX on larvae of the non-biting midge *C. riparius* in sediment-water test systems at different temperatures. Whereas the first set of experiments, performed under continuous exposure (with steady natural degradation of TMX), was used to calibrate the DEB-TKTD model, a second set of experiments, using pulsed exposures, was used to validate the model.

2.1.1. Test substance

The test substance thiamethoxam (TMX) was provided by Syngenta Crop Protection, LLC (Greensboro, NC, USA) in solid form with 99 % active ingredient content. A stock solution was prepared by weighing 25.2 mg TMX on an analytical balance, transferring the substance to a 100 mL volumetric flask, and filling it to the 100 mL mark with freshly prepared and aerated artificial Elendt M4 medium (Elendt, 1990). This stock solution was stirred at room temperature until fully dissolved. A sub-stock solution of 2.5 mg/L was prepared by 1:100 dilution with M4 medium. The sub-stock solution was further diluted to achieve the desired test concentrations in the test systems as described below.

2.1.2. Test organisms

All tests were performed with larvae collected from an in-house culture of *C. riparius*. The cultures were reared at 20 °C and a light-dark regime of 16:8 hours. The artificial Elendt M4 medium was used as a medium for both the cultures and the experiments. 3–4 days before the start of each test, all egg masses present in the culture chambers were removed. This was done to ensure that on the next day, only newly laid egg masses (< 1 d) were present, which were then collected and cultivated in M4 medium until hatching. The first instar larvae (< 12 h) were then added to the respective test systems.

2.1.3. Test systems

The sediment used in the experiments was formulated according to OECD guideline 219 (OECD, 2023). The individual components were mixed 7 days before the test started, and the pH was adjusted to a range of 6.0–9.0. 105 g of the formulated sediment was added to each glass beaker (volume: 600 mL, diameter: 8 cm), forming a sediment layer of ~1.5 cm. Next, 250 mL of M4 medium was added to each beaker. The sediment-water test systems were kept under test conditions with continuous aeration for 7 days. At the test start, the aeration was turned off, and 20 larvae per beaker were added in a randomized order. 24 hours later, the medium was spiked by evenly mixing the appropriate amount of TMX stock solution into the water phase to achieve the desired (initial) test concentration in each beaker, after which the aeration was turned on again.

Larvae were fed three times a week with a suspension of 1 g Tetra Phyll in 20 mL deionized water. A food quantity of 0.4 mg per larva per day was given for the young larvae, whereas this amount was increased to 0.75 mg later in the experiment when larvae were further developed (i.e., after 10 d in all test setups except those performed at 15 °C and 12 °C, where the amount was increased only after 15 and 20 days, respectively). All experiments were performed with a light intensity of 500–1000 lux and a light-dark regime of 16:8 hours. Medium losses due to evaporation were compensated once a week with deionized water. Water temperatures were continuously monitored and recorded with a data logger (1 h interval). M4 medium renewals were only done in the test setups using pulsed exposure when an abrupt removal of the test chemical was desired.

2.1.4. General test design: development and reproduction

To assess the midges' development time until emergence, 4 replicates of the sediment-water test system were prepared for each

treatment group and left undisturbed throughout the test. From the time when emergence could be expected, the test systems were examined daily for emerged individuals. Upon emergence, adult midges were removed from the test vessels, and the number of emerged individuals was recorded. A minimum emergence rate of 70 % in the control at the test end was used as a validity criterion. To assess reproduction, emerged females from the same test were individually matched with males from the culture at a ratio of 1:3 at 20 °C. Whenever possible, a total of 20 females were collected from each treatment group and placed into individual 400 mL glass beakers containing M4 medium of ~1 cm depth, after which three male midges per female were added. The glass beakers were covered with gauze and kept under culture conditions. The egg masses deposited in the beaker were collected and individually transferred to small glass jars containing M4 medium. The number of eggs per egg mass was counted under a stereo microscope for each treatment group in 5 egg masses.

2.1.5. General test design: growth and survival

To assess the midges' growth and survival, 5–10 replicates (depending on the test) of the sediment-water test system were prepared for each treatment group. The growth and survival of the larvae over time were determined using destructive sampling of individual replicates. During the destructive sampling, a test vessel was completely emptied, and its sediment was spread out on a large petri dish. For each treatment group and time point, one single replicate was examined. The larvae's survival was assessed using immobility as a proxy, i.e., if a larva did not respond to a mechanical stimulus, it was considered dead. For larval growth, the mean dry weight of all surviving individuals per test vessel was determined. To this end, all living individuals (larvae + pupae) of one replicate were pooled, dried at 65 °C for 24 h, and weighed.

2.1.6. Test details: calibration experiments using continuous exposure

The first set of experiments was performed to generate data to calibrate the DEB-TKTD model. Four chronic sediment-water toxicity tests were conducted, each at a different temperature, during which the test organisms were exposed to one of 5 concentrations (12 °C: 18–46 µg/L; 15 °C: 11–39 µg/L; 20 °C: 4–32 µg/L; 23 °C: 4–32 µg/L) or a control treatment. Note that at higher temperatures, lower concentrations were used because preliminary tests had already indicated a toxicity-enhancing effect at those temperatures, and we aimed to generate effect data preferably at several concentrations with a mid-range effect to calibrate the model with. Natural degradation of TMX was not compensated for. Although this led to a dynamic decline of the concentration over time, this was taken into account in the subsequent modeling (see methods section *TKTD model calibration*). For each temperature-concentration combination, 4 replicates were prepared for the development and reproduction part of the test, whereas 10 replicates were prepared for the part assessing growth and survival (7–8 of these were evaluated while the remaining served as backups). The sampling times for the destructive samplings were based on pre-tests (performed at the same four temperatures) and were planned in such a way that 6–7 samplings could be completed before most of the larvae would emerge at the respective temperature. Analytical samples of the water phase were taken 1 h after spiking. Additional analytical samples of both the water and sediment phase were taken halfway through and at the end of the experiment, to assess the degradation of TMX and potential losses from the water phase to the sediment over time. A detailed overview of the nominal initial concentrations per temperature treatment, and associated sampling times can be found in the supporting information, section S1, Table S1.

2.1.7. Test details: validation experiments using pulsed exposure

The second set of experiments was performed to generate data to test the DEB-TKTD model under new and slightly more complex conditions. Two different temperature scenarios were tested with two pulse

exposure patterns each. While the first temperature scenario represented a constant temperature (CT) of 20 °C, the second scenario included a dynamic temperature (DT), gradually increasing from 15 to 23 °C in steps of 1 °C every 2–3 days. The exposure patterns consisted of two pulses each, either with a relatively long no-exposure phase between the pulses allowing for potential damage repair (DR) within the organisms before the second pulse, or with only a short no-exposure phase, presumably causing more stress from the first pulse to be carried over (CO) – and added to – that of the second pulse. Whereas both pulses had different lengths of 2 and 3 days respectively, they were spiked with the same initial concentration, i.e., 30 µg/L or 60 µg/L, or 0 µg/L as a control treatment. The exact timing of the pulses is presented in Table S2 (supporting information, section S1). Following the abbreviations provided above, the four temperature-exposure patterns are hereafter named CT-DR, CT-CO, DT-DR, and DT-CO. As in the calibration experiments, the natural degradation of TMX during the duration of a pulse was not compensated. At the end of each pulse, the medium was carefully replaced (~95 %) with fresh uncontaminated medium to start the post-exposure period. For each temperature-exposure pattern combination, 4 replicates were prepared for the development and reproduction part of the test. Since no significant effects on the egg mass size (neither by temperature nor by the chemical) had been found in the calibration experiments, only the development time until emergence was assessed as an endpoint. 5 replicates were prepared for the growth and survival part of the test (four of these were evaluated while one served as a backup). The sampling times for the destructive samplings were based on preliminary model simulations and were planned so that four samplings could be completed before most of the larvae would emerge. Analytical samples of the water phase were taken 1 h after spiking and at the end of each exposure pulse. Additionally, water samples were taken one day after a pulse had been terminated by ~95 % renewal of the medium. A detailed overview of the temperature-exposure patterns assessed in the validation experiments, and associated sampling times can be found in Table S2.

2.1.8. Chemical analysis

Analytical samples of the water phase of each test treatment were created by pooling water column samples (10 mL each) from its individual replicates. 4 mL of the pooled sample were stabilized with 1 mL of acetonitrile and stored at –18 °C. For sediment sampling, at first, all larvae were carefully removed from the sediments. Individual replicates of each treatment were then pooled, transferred to 150 mL glass bottles, and stored at –18 °C. The chemical analysis of the test samples of both the water and sediment phases was performed by ibacon GmbH (Roßdorf, Germany) using LC-MS/MS technique. A detailed description of the analytical method can be found in the supporting information, section S2, Table S3.

2.2. Mechanistic evaluation using a bioenergetics-based life-history model for *C. riparius*

In order to extrapolate our laboratory findings to realistic field conditions, we parametrized a DEB model that includes a TKTD sub-model quantifying chemical-induced effects. A detailed introduction to the DEB theory and the standard DEB model has been presented elsewhere (Kooijman, 2009, 2010). Here, we focus particularly on the DEB model variant for holometabolous insects (Llandres et al., 2015; Kooijman, 2010) and on assumptions made specifically for this study, especially with respect to the TKTD model extension and temperature dependencies.

2.2.1. Physiological model with temperature dependence

In a first step, it was ensured that the life history of *C. riparius* at unstressed conditions, as observed in our experiments, could be adequately reproduced by the core model. An existing Add-my-Pet (AmP) database entry for *C. riparius* (Augustine and Gergs, 2019) was

used as the starting point for model calibration. Unfortunately, when familiarizing with the applied model variant *hax*, an error in the equations linking observations to predictions (for this model type) in AmP's parameter estimation tool *DEBtool* was identified, making the existing parameter set unusable. While the error was quickly acknowledged and corrected by the curators, a re-parametrization of the *hax* model for *C. riparius* was still needed. The decision was made to re-parametrize the model with both the data from our new experiments and existing data from the AmP entry. The corrected version of *DEBtool* was used for the parameter estimation in [MATLAB \(2017\)](#); see supporting information, section S3 for further details. A complete overview of the used model

Table 1

DEB-TKTD model parameters used in this study. The physiological DEB parameters of the *hax* model were optimized to reproduce control life-history data of *C. riparius* at different temperatures. Note that some parameters were fixed to commonly used default values as they cannot be reliably estimated from the data typically available from most life-history studies. TKTD parameters were optimized to reproduce concentration-dependent lethal and sublethal effects of TMX in *C. riparius*. The sublethal effect parameters assume an effect on feeding via the functional response f .

Symbol	Unit	Interpretation	Estimate
Physiological DEB parameters			
$\{F_m\}$	$L d^{-1} cm^{-2}$	Specific searching rate	6.5 (fixed)
κ_X	—	Assimilation efficiency	0.8 (fixed)
$\{P_{Am}\}$	$J d^{-1} cm^{-2}$	Surface-area-specific maximum assimilation rate	816.20
$\dot{v}$	$cm d^{-1}$	Energy conductance	0.0154
κ	—	Allocation fraction to soma	0.400
κ_R	—	Reproduction efficiency	0.95 (fixed)
$\{P_M\}$	$J d^{-1} cm^{-3}$	Volume-specific somatic maintenance rate	3474
$\{P_T\}$	$J d^{-1} cm^{-2}$	Surface-area-specific somatic maintenance rate	0 (fixed)
k_J	d^{-1}	Maturity maintenance rate coefficient	0.002 (fixed)
$[E_G]$	$J cm^{-3}$	Volume-specific costs for structure	4400 (fixed)
E_H^b	J	Maturity level at birth	5.31×10^{-3}
E_H^p	J	Maturity level at puberty	1.74×10^{-2}
E_H^e	J	Maturity level at emergence	3.79×10^{-3}
$[E_R]_j$	$J cm^{-3}$	Reproduction buffer density at pupation (independent of f in the default <i>hax</i> model but assumed to be only true at $f = 1$ in this study)	3.54×10^4
$(E_R/L)_j$	$J cm^{-1}$	Reproduction-buffer-to-structural-length ratio at pupation (assumed to be independent of f in this study)	195.08
κ_V	—	Energy conversion efficiency from reserve to structure (during growth) and then back to reserve (at pupation)	0.7 (fixed)
T_{REF}	K	Reference temperature	293.15 (fixed)
T_A	K	Arrhenius temperature	7377
T_H	K	Upper boundary temperature	293.6
T_{AH}	K	Arrhenius temperature for upper boundary	5949
s_{shrink}	—	Shrinking stress coefficient	4.13
TKTD parameters			
k_d	d^{-1}	Dominant rate constant	8.10×10^{-2}
T_{Akd}	K	Arrhenius temperature for the dominant rate constant	1.65×10^4
z_b	$\mu g L^{-1}$	Effect threshold sublethal	6.39×10^{-2}
b_b	$L \mu g^{-1}$	Effect strength sublethal	3.30×10^{-2}
z_s	$\mu g L^{-1}$	Effect threshold survival	1.40
b_s	$L \mu g^{-1}$	Effect strength survival	1.65×10^{-2}
h_b	d^{-1}	Background hazard rate	1.33×10^{-6}

parameters with corresponding interpretations and units is given in [Table 1](#). All calibration files used in this study, including the calibration data, can be accessed on GitHub via: https://github.com/gaiac-eco/2024_C_riparius_TMX_DEB-TKTD.

A full model description of the *hax* DEB model variant for holometabolous insects can be found in the supporting information, section S4. In short, just like the standard DEB model (*std*), the *hax* model follows the 4 state variables energy reserve E (J), body structure V or L^3 (cm^3), maturity E_H (J), and reproduction buffer E_R (J) over time. The reserve is filled through energy uptake from food, and is remobilized to fuel all energetic needs within the organism, including energetic costs for the build-up and maintenance of structure, the build-up and maintenance of maturity, and once fully matured, the energetic investment in the reproduction buffer. The *hax* model deviates from the *std* model in that the animal experiences a phase of metabolic acceleration (and accelerated growth) between the onset of feeding and the start of investment in reproduction. In DEB terminology, these two events are referred to as *birth* and *puberty*, respectively. While no further life-history events occur after puberty in the *std* model (except for regular releases of eggs/neonates), *hax* animals undergo the additional events of pupation and emergence. Pupation – according to the documented *hax* model assumptions at the time of writing ([Kooijman, 2010](#), update 2023/05/05) – occurs as soon as a critical threshold of reproduction buffer density E_R/V (J/cm^3) is reached. At this point, all body structure is transformed into reserve energy, and maturity is reset to zero. The pupa does not feed externally but draws from its reserve, much like the embryo (i.e., pre-birth) stage of the DEB models. Emergence is triggered by a threshold of maturity. The imagoes, who still do not feed, finally transform their reproduction buffer into a single egg mass and are assumed to die shortly after. One important modification to the *hax* model was made in this study: it was assumed that pupation is triggered by a critical threshold of the ratio between the reproduction buffer E_R and the structural length L (cm) rather than the ratio between E_R and the structural volume L^3 . This decision was made after noting that the original version led to a drastically premature pupation (and emergence) when modeled feeding conditions were poor (or when assimilation was impaired by a stressor), the opposite of which was shown to be true in laboratory experiments ([Péry et al., 2002](#)). Alternative assumptions that were considered in this study included a critical threshold for L (absolute), E_R (absolute), or the ratio between E_R and L^2 as theoretical triggers for pupation, though L/E_R appeared the most reasonable. For more details, refer to the supporting information, section S5.

To account for temperature dependence of the modeled processes, a temperature correction factor TC (—) was used, which is consistently applied to all physiological rates (i.e., all parameters with dimension $1/d$):

$$TC = \begin{cases} \exp\left(\frac{T_A}{T_{REF}} - \frac{T_A}{T}\right), & T < T_{REF} \\ s_H \exp\left(\frac{T_A}{T_{REF}} - \frac{T_A}{T}\right), & T \geq T_{REF} \end{cases} \quad (1)$$

with the upper boundary correction s_H (—):

$$s_H = \frac{1 + \exp\left(\frac{T_{AH}}{T_H} - \frac{T_{AH}}{T_{REF}}\right)}{1 + \exp\left(\frac{T_{AH}}{T_H} - \frac{T_{AH}}{T}\right)} \quad (2)$$

Below the reference temperature T_{REF} (K), the temperature dependence is described by only two parameters: T_{REF} and the Arrhenius temperature T_A (K). At and above T_{REF} , two additional parameters, i.e., the upper boundary temperature T_H (K) and the Arrhenius temperature for the upper boundary T_{AH} (K), are used to model a flattening of the temperature curve where physiological tolerance is exceeded. The decision to include the upper boundary correction was mainly made because it was also included in the existing AmP entry for *C. riparius* and

this extension of the basic two-parameter Arrhenius formulation delivered a slightly superior model fit to the data (see supporting information, section S6).

Finally, it should be noted that submodels accounting for age-related mortality, which are often considered in DEB parametrizations, were ignored in this study since (a) age mortality in holometabolous insects is tightly linked to specific life-history events, something that is not accounted for in DEB's aging submodel by default and (b) the dynamics of age-related mortality were of minor relevance for this study, which focused primarily on the larval stage (for which a background mortality was parametrized in the TKTD part of the model).

2.2.2. TKTD model extensions

To describe both lethal and sublethal effects in a toxicokinetic-toxicodynamic (TKTD) framework, we used established DEB-TKTD formulae as specified by Jager et al. (2023), with some minor modifications worked out iteratively (see discussion and supporting information, section S7) to capture all patterns observed in the experimental data from the calibration experiment adequately.

The toxicokinetics were described by a one-compartment model for the scaled internal damage D ($\mu\text{g/L}$), which captures all dynamic processes of damage accrual and repair. The dynamics of D are often assumed to – but do not necessarily – follow directly from the uptake and elimination kinetics of the active substance in the organism. Since damage is a rather abstract state variable, which cannot be measured directly, it is scaled with – and expressed in the unit of – the external concentration of the substance c ($\mu\text{g/L}$). Commonly used feedbacks from the physiological model to the damage dynamics, i.e., a size correction accounting for changes in the surface-to-volume ratio, and a term accounting for *dilution by growth*, were considered but ultimately excluded from the model due to an inferior fit to the experimental data.

$$\frac{d}{dt}D = k_d(c - D) \quad (3)$$

Effects on the survival of an individual are modeled following the *stochastic death* assumption (Jager et al., 2011). The probability of an individual to die is given by the hazard rate h (1/d), the sum of a constant background hazard rate h_b (1/d), the starvation hazard rate h_s (see supporting information, section S4), and a damage-dependent hazard rate h_c (1/d):

$$h = h_b + h_s + h_c \quad (4)$$

with

$$h_c = b_s \max(0, D - z_s \frac{L_m}{L}) \quad (5)$$

and

$$L_m = \kappa \frac{\{\dot{p}_{Am}\}}{\{\dot{p}_M\}} \quad (6)$$

The non-standard multiplication term $\frac{L_m}{L} (-)$ in Eq. 5 accounts for a size-dependent sensitivity via the lethal effect threshold z_s ($\mu\text{g/L}$) deduced from the data on survival over time in the calibration experiment (see discussion and supporting information, section S7).

Sublethal effects were modeled using a stress function s ($-$), which above a damage threshold z_b ($\mu\text{g/L}$) increases linearly with D . The associated slope is given by the effect strength b_b ($\text{L}/\mu\text{g}$):

$$s = b_b \max(0, D - z_b) \quad (7)$$

This stress function is typically assumed to act on a single model parameter p following:

$$p = p_0(1 + s) \quad (8)$$

or

$$p = p_0 \max(0, 1 - s) \quad (9)$$

Here, p_0 is the parameter at unstressed conditions. Whereas different physiological parameters can be targeted by the stress function (following different so-called *physiological modes of action* or *pMoAs*; Álvarez et al., 2006), in this study, it was found that an effect on feeding (i.e., ingestion or assimilation rate) via the functional response f delivered the best results:

$$f = f_0 \max(0, 1 - s) \quad (10)$$

A comparison between this and the second most plausible pMoA, i.e., a pMoA on the costs for growth, can be found in the supporting information, section S8.

For the TKTD part of the model, it was assumed that temperature dependence in the toxicokinetics (via the dominant rate constant k_d , $1/d$) can be described using the same formula used for the physiological processes, although with a separately calibrated Arrhenius temperature T_{Akd} (K) and no upper boundary. This was done because the upper boundary is meant to account for temperatures at which physiological processes are impaired due to temperature stress, which – albeit debatable – should not be the default assumption for the toxicokinetics one-compartment model of damage accrual and repair (or chemical uptake and elimination, especially when these are assumed to be passive processes). No temperature correction was applied to the lethal effect strength parameter b_s , or any of the remaining TKTD parameters; this is further elaborated in the discussion and the supporting information, section S6.

2.2.3. TKTD model calibration

The estimation (or *calibration*) of the TKTD parameters, like the control model, was done in DEBtool. In this step, all of the physiological model parameters, that had previously been fitted to the control data, remained fixed, whereas all TKTD parameters and their temperature correction terms were estimated simultaneously in the DEBtool's fitting routine. Model simulations recreating the calibration experiments were carried out in such a way that they reflected the experimental conditions as closely as possible to ensure a correct parametrization. An observed decline of TMX in the water phase over time (due to degradation and losses to the sediment) was reproduced using a simple temperature-dependent degradation kinetics model (presented in the supporting information, section S2) calibrated with the analytical data. This model was used to interpolate a time-dynamic concentration profile for each individual model simulation, using the spiking time, spiking concentration, and mean measured temperature as inputs. Since experimental temperature fluctuations were very low, mean measured temperatures were used in the simulations. Furthermore, it was assumed that the food provided in the experimental setup corresponded to *ad libitum* feeding conditions at all times, resulting in a functional response of $f = 1$, except when organisms were impacted by the sublethal pMoA on feeding as detailed above. While the model's aim is to describe an average individual, due to biological variability, not all larvae pupate at the same time. Therefore, some individuals remained in the larval stage after the point where the model already suggested a growth stop due to pupation. The pupation life-history event was turned off to include such growth and survival data in the calibration. The normalized root mean squared error (NRMSE) was used as a quantitative criterion to rate the model fits. A value of 0.5 was proposed as a general upper acceptable limit for survival predictions by EFSA (2018), although the authors refrained from setting strict cut-off criteria, and predictions of sublethal endpoints may need to be judged differently.

2.2.4. Model validation simulations

Model simulations recreating the validation experiments using the four different temperature-exposure patterns CT-DR, CT-CO, DT-DR, and DT-CO, were performed following the same paradigm of *as-realistic-as-possible*. Concentration profiles were computed considering the timing of spikings and medium renewals, measured spiking

concentrations, and the degradation kinetics of TMX in the water phase as calibrated with analytical data from the calibration experiments. This time, real measured temperature profiles were read into the model, since the DT-scenarios were partially dynamic. The life-history event of pupation was again turned off for the same reason described above. The NRMSE was used to assess the goodness of fit.

2.2.5. Simulations of realistic exposure scenarios

In a final step, the model was used to simulate the chironomids' life cycle at realistic exposure conditions using two reference scenarios measured in streams in Missouri and Nebraska watersheds. These two watersheds are part of the Atrazine Ecological Monitoring Program (AEMP), an ongoing, multi-year environmental monitoring program with near daily water surface sampling. The AEMP was designed to target the most vulnerable headwater streams to atrazine exposures in the main product use regions (USEPA, 2006, 2020), and also monitors water concentrations of other pesticides, such as TMX. Since the chironomids' life cycle is much shorter than the reference scenarios, a moving time window (MTW) approach was used, where life-cycle simulations were repeatedly performed, each time starting at a different,

incrementally shifted day of the respective exposure scenario (EFSA, 2018; Sherborne et al., 2020). Whereas so far, MTW simulations with DEB-TKTD models were mostly done using fixed temperatures, thereby simulating a standard chronic test, here we used the measured temperature profiles belonging to each scenario. Simulations at a fixed temperature of 20 °C were done for comparison. The length of the time window, at the end of which lethal and sublethal effects are registered, typically also matches that of a standard chronic test. The standard test used in this study, however, is unique in that endpoints such as survival and growth are only followed up until emergence occurs. Therefore, instead of a fixed time window length, we used the moment of pupation (which occurs shortly before emergence) as the time point to register effects and end the time window. Compared to a fixed MTW length, this approach also provides more easily interpretable results for simulations with dynamic temperatures since effects of temperature on the developmental rate (and thus length of the virtual test) are accounted for. An in-house automated routine (inspired by the suggestions made by Sherborne et al., 2020) was used to determine EP₁₀ values, i.e., the minimum exposure multiplication factors leading to a 10 % effect strength (10 % reduction in survival or dry weight) at the end of each

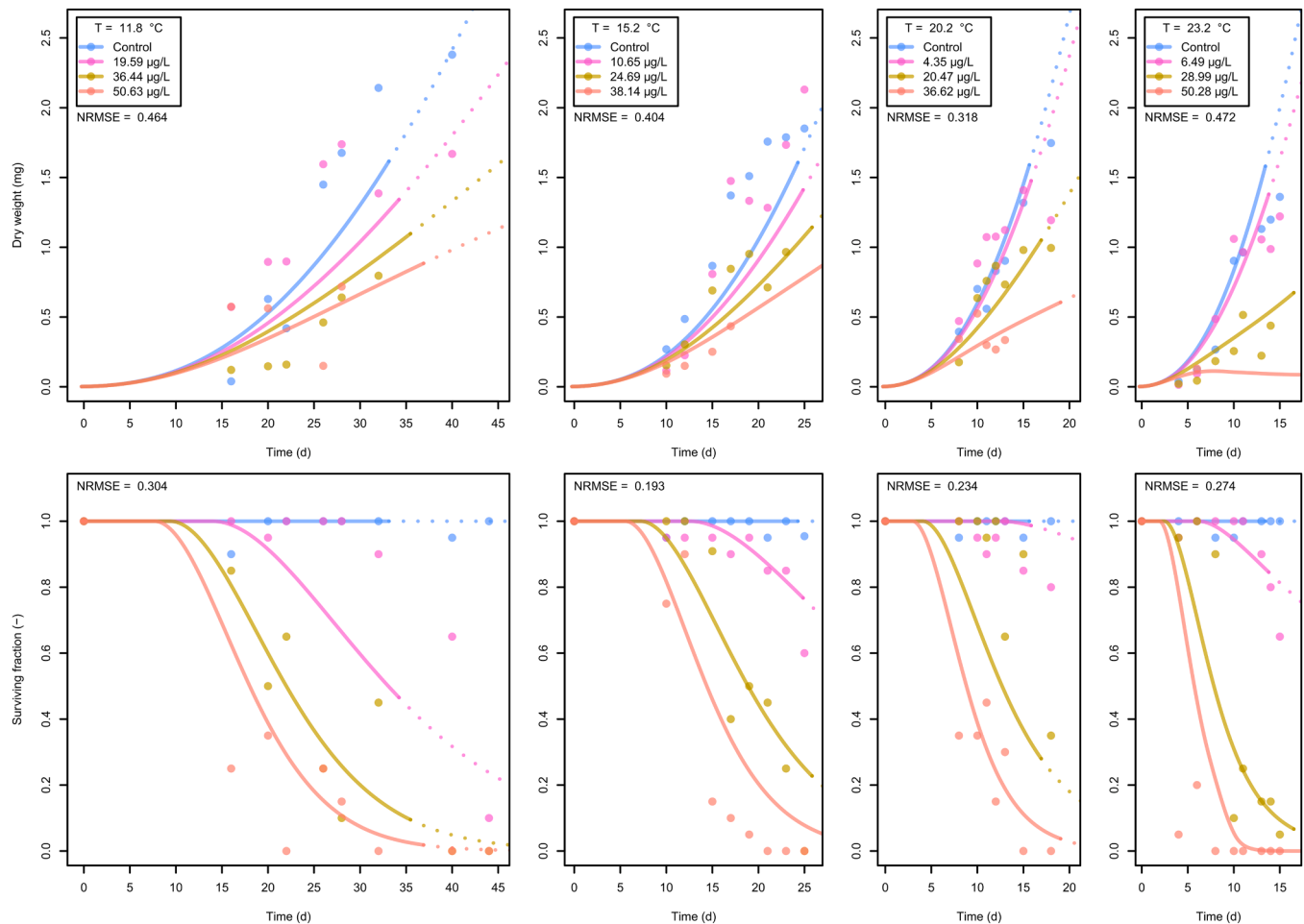


Fig. 1. Growth and survival data from the calibration experiment and corresponding model simulations for an *in silico* recreation of the experimental conditions. The physiological model had been parametrized with the control data only (and control data from other studies found in the AmP database) whereas the TKTD-parameters and their temperature dependencies had been subsequently fitted to the treatment-related data. Note that not all data used in the respective parametrization steps are represented in this figure. Each column represents one of the four temperature setups used in the calibration experiments. Simulated data are presented as lines whereas measured data are presented as filled circles. The transition from a solid to a dotted line indicates the point at which, according to the model, pupation occurs in an average individual. Since the experimental data beyond this point were still obtained from larvae (some developing more slowly due to natural variation), pupation had been turned off for the simulation. Indicated test concentrations correspond to analytically measured concentrations of thiamethoxam 1 h after spiking. Note that the same color can represent different concentrations depending on the temperature setup. For better clarity, only three of the five concentration levels (i.e., the lowest, medium, and highest test concentrations) per temperature setup are shown. The normalized root mean squared error (NRMSE) presented for each panel also incorporates the data of the intermediate concentration levels.

window (at pupation). For the most adverse MTW, i.e., the one with the lowest EP_{10} , additional EP_x values for effect strengths up to 90 % (EP_{90}) were computed in 10 % steps.

3. Results

3.1. Calibration data

Detailed results of the chemical analysis for the calibration experiments are provided in the supporting information, section S2, Table S4. In the temperature scenarios at 12, 15 and 20 °C, the analytical results showed a recovery of 93–115 % of nominal for the initial concentrations in the water phase 1 h after spiking, whereas for 23 °C, the recovery was 156–164 %, indicating that exposure concentrations in this scenario were higher than intended. The mean measured water temperatures ($\pm$ standard deviation) during the calibration experiments were 11.8 ± 0.5 °C, 15.2 ± 0.6 °C, 20.2 ± 0.3 °C, and 23.2 ± 0.5 °C.

Data on growth and survival over time in the calibration experiments are presented in Fig. 1. Note that for better clarity, only three of the five tested concentration levels (i.e., the lowest, medium, and highest test concentrations) per temperature setup are presented. An analogous figure, also including the intermediate concentrations, can be found in the supporting information, section S7, Figure S5. The data clearly indicate a faster growth at higher temperatures within the range of tested temperatures. At the same time, no noticeable effects on survival occurred at any of the different temperatures' control treatments, hinting at a very low background mortality and indicating that no direct effects of temperature on survival were present in this experiment. Furthermore, concentration-dependent effects of TMX were observed for both endpoints. Firstly, the data show a tendency toward lower mean

dry weights per observation time at higher exposure concentrations, a trend that became more pronounced over time. Secondly, higher concentrations of TMX led to a faster decrease in the number of surviving individuals over time. It is also worthwhile noting that the timing of the onset of first effects on survival appeared to be concentration-dependent, with effects occurring later at lower test concentrations. At the highest test concentrations (regardless of the temperature setup), no dry weight measurements could be performed toward the end of the test since not enough midges had survived at that point. Moreover, especially with regard to the survival data, higher temperatures accelerated the effects of TMX on the survival of *C. riparius* as similar concentrations led to earlier and stronger effects on survival at higher temperatures (e. g., 95 % survival at 19.6 $\mu\text{g/L}$ and 12 °C on day 20 versus 35 % survival at 20.47 $\mu\text{g/L}$ and 20 °C on day 18).

Results from the development and reproduction part of the calibration test can be found in Fig. 2. Additionally, cumulative emergence curves are presented in the supporting information, section S9. Firstly, it can be noted that the presented endpoints, i.e., the time until emergence, and the number of eggs per egg mass, could only be obtained for individuals that were able to emerge and reproduce successfully, which was severely hindered by mortality effects and possibly additional developmental effects at the higher test concentrations. Hence, concentration-dependent effects on these endpoints – if present – could only be analyzed for the lower end of the tested concentration range. Time until emergence data were analyzed using a Kruskal-Wallis test ($\alpha = 5$ %) followed by Dunn's test for pair-wise comparison ($\alpha = 5$ %, R package 'dunn.test'; Dinno, 2017) in R (Ver. 4.2.1; R Core Team, 2022). The time until emergence showed a decrease with increasing temperatures which was significant for each comparison between two different temperatures, independent of exposure concentration. Furthermore, no

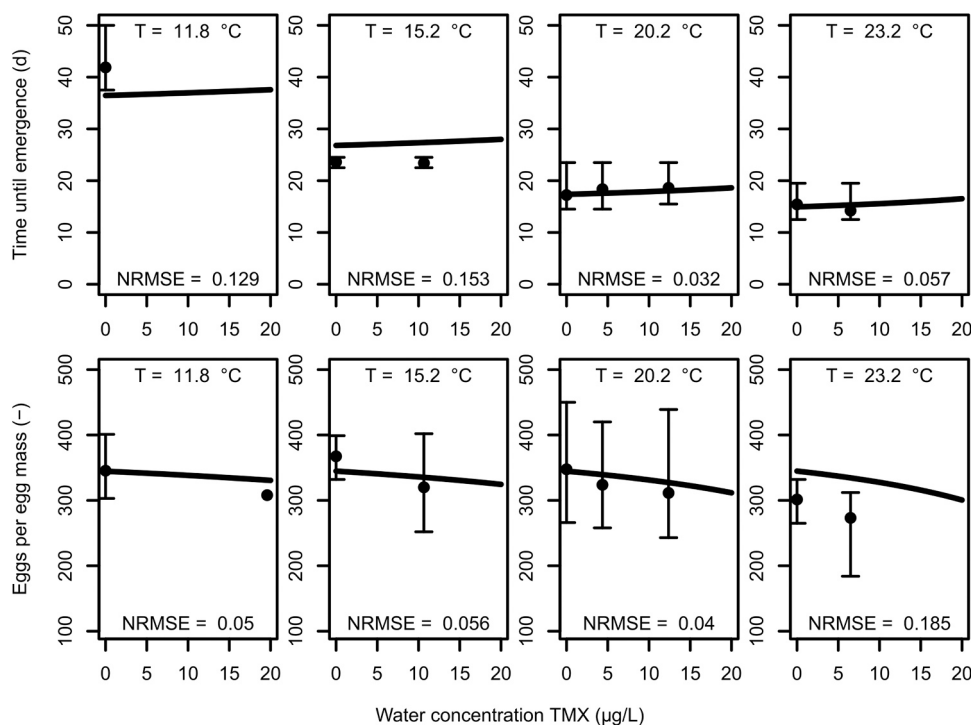


Fig. 2. Time-untill-emergence data and reproduction data expressed as number of eggs per egg mass in the calibration experiment with corresponding model simulations for an *in silico* recreation of the experimental conditions. The core model had been parametrized with the control data only (and control data from other studies found in the AmP database) whereas the TKTD-parameters and their temperature dependencies had been subsequently fitted to the treatment-related data. Note that only a fraction of the data used in the respective parametrization steps are represented in this figure. Each column represents one of the four temperature setups used in the calibration experiments. Simulated data are presented as solid lines whereas measured data are presented as filled circles (mean) with whiskers (range). Indicated test concentrations correspond to analytically measured concentrations of thiamethoxam 1 h after spiking. Note that data were only available for the lower range of the tested concentrations since not enough individuals were able to successfully emerge at the higher concentrations. Normalized root mean squared errors (NRMSE) are presented as a goodness-of-fit metric.

significant effect of exposure to TMX was observed for any of the temperature configurations within the limited range of available data. Regarding the number of eggs per egg mass, a tendency toward smaller egg masses at higher TMX concentrations, and a slightly impaired egg production at 23 °C, was observed, though no significance was found in a one-way ANOVA ($\alpha = 5\%$) over all available temperature-concentration combinations.

3.2. Calibration simulations

The physiological parameters and the TKTD parameters of the calibrated model can be found in Table 1. Results of the model simulations, recreating the calibration experiments, are presented together with the experimental data in Figs. 1 and 2. NRMSE scores were all below 0.5, i. e., the general upper acceptable limit suggested by EFSA (2018) when dealing with survival data.

The model underpredicted the growth in the control and the lowest

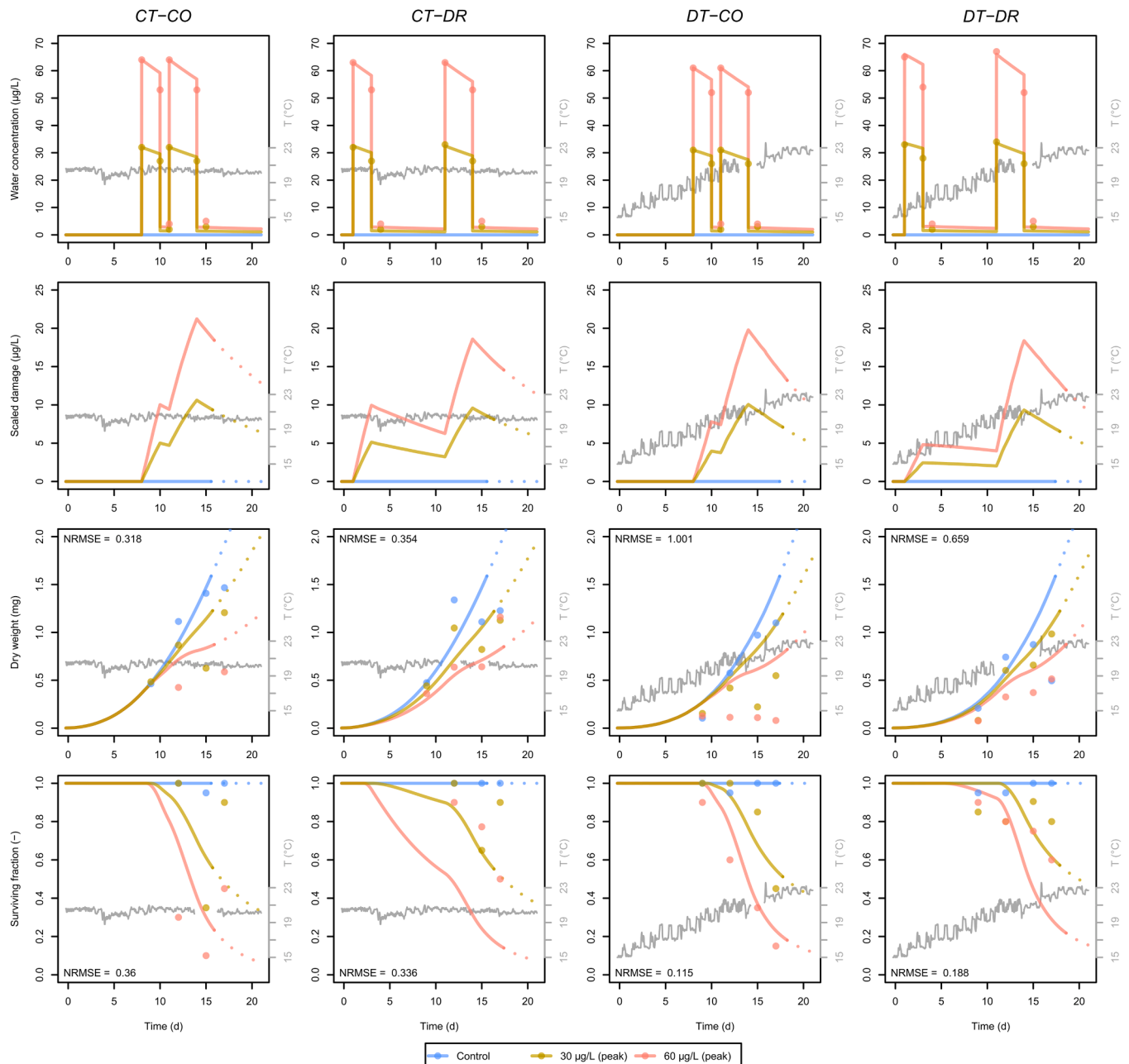


Fig. 3. Growth and survival data from the validation experiment and corresponding model simulations for an *in silico* recreation of the experimental conditions. Each column represents one of the four temperature-exposure pattern combinations applied in the validation experiment, each consisting of two-pulse exposures with either narrow or wide pulse spacing (as indicated by the top panels) and either a stable (20.3 ± 0.3 °C, mean $\pm$ SD) or an increasing (15–23 °C) temperature (as indicated by the overlaid temperature profiles scaled to the secondary y-axes). Mean dry weights and survival over time are presented alongside the concentration profiles of thiamethoxam in the water phase and the simulated scaled internal damage in the organisms. Simulated data are presented as solid lines whereas measured data (except for the temperature profiles) are presented as filled circles. The transition from a solid to a dotted line indicates the point at which, according to the model, pupation occurs in an average individual. Since the experimental data beyond this point were still obtained from larvae (some developing more slowly due to natural variation), pupation had been turned off for the simulation. The measured temperature profiles are partly hidden where they might otherwise hinder readability of the other data. Normalized root mean squared errors (NRMSE) are presented as a goodness-of-fit metric.

TMX concentration in the two coldest treatments, whereas no clear tendency to over- or underprediction of growth effects related to the different TMX levels could be observed, partly because the experimental data had a relatively high range of variation.

After some iterative model refinements (see supporting information, section S7), two qualitative observations in the survival data, (a) the concentration-dependent delay in the occurrence of first mortality effects, and (b) the occurrence of effects also at the lowest test concentrations, were successfully reproduced. Effects on survival were slightly underpredicted in the highest TMX concentration levels at all four temperatures. Furthermore, effects on survival in the lowest TMX concentration were overpredicted at 12 °C (except at test end), whereas they were underpredicted at 20 °C.

When considering development times, besides a clear temperature dependence in the time until emergence (longer development times at lower temperatures, reflected in both the data and simulation results), the model also predicted a slight increase in this endpoint with increasing concentrations of TMX. This observation, however, could not be statistically verified with the limited experimental data available for this endpoint. Another subtle but non-significant trend toward fewer eggs per egg mass at higher concentration levels of TMX was qualitatively represented by the model. Furthermore, by design (following from the default assumptions of the DEB theory), the model predicted no effect of temperature on the number of eggs per egg mass in the unexposed control treatments, which is also supported by the data.

3.3. Validation data

Detailed results of the chemical analysis for the validation experiments are provided in the supporting information, section S2, Table S5. The results showed a recovery of 102–113 % of nominal for the initial pulse concentrations in the water phase 1 h after spiking over all scenarios. The mean measured water temperature ($\pm$ standard deviation) of the CT scenarios was 20.3 ± 0.4 °C. Detailed temperature profiles for the validation experiments are presented in Fig. 3.

Data on growth and survival over time in the validation experiments are presented in Fig. 3. Overall, the validation test results qualitatively indicate similar concentration-response relationships for both endpoints, as observed in the calibration experiments. Results from the development part (without reproduction) of the validation experiments are presented in Fig. 4. Unfortunately, for unclear reasons, one of the four test scenarios (DT-CO) in the development test had to be excluded from evaluation since the fraction of successfully emerged individuals in the control at test end (23 %) was far below the cutoff limit of 70 % for this sub-experiment to be considered valid. The three remaining

temperature-exposure pattern combinations all showed a significantly slower development at pulses of 30 $\mu\text{g/L}$ compared to the control (Kruskal-Wallis test followed by Dunn's test; $\alpha = 5$ %), whereas no data could be obtained at 60 $\mu\text{g/L}$ per pulse where only up to 2.5 % (throughout all three scenarios) of the midges had successfully emerged.

3.4. Validation simulations

The simulations of the dry weight and the survival over time, recreating the validation experiments, are presented together with the experimental data in Fig. 3. Overall, the model predicted survival in different scenarios better than it did growth, as demonstrated by overall lower NRMSE scores. The model predicted survival under dynamic temperature (NRMSE = 0.115 and 0.188) better than it did under constant temperature (NRMSE = 0.360 and 0.336). However, growth predictions were best under constant temperature scenarios (NRMSE = 0.318 and 0.354 versus 1.001 and 0.659). There was some underprediction of effects on growth, especially in the highest treatments of the CO scenarios; the model mostly overpredicted effects on survival in both DR scenarios. Except for the growth simulations under DT, NRMSE values were all below 0.5.

Simulation results for the development time until emergence in the validation experiments are shown in Fig. 4. For the limited data available (mostly because of the high mortality in all treatments with 60 $\mu\text{g/L}$ per pulse), the model simulations qualitatively captured the concentration-dependent developmental delay seen in the data. NRMSE values were all below 0.5.

3.5. Simulations of realistic exposure scenarios

Using the profiles of measured water temperatures for the two reference scenarios (Fig. 5), minimum EP_{10} values for the survival at pupation were found at 21.87 for the Missouri scenario (MTW starting day 2021/06/19), and at 44.64 for the Nebraska scenario (MTW starting day 2021/05/28). When a constant temperature of 20 °C was used instead (supporting information, section S10, Figure S12), a higher EP_{10} of 25.67 was obtained for the Missouri scenario (MTW starting day 2021/06/19), whereas a lower value of 38.58 was found for the Nebraska scenario (MTW starting day 2021/05/27). A similar quantitative offset of EP_x values between the simulations, with dynamic temperatures versus the constant temperature, was also observed for all other effect strengths between 10 % and 90 %, as shown in Fig. 6. Analogous results of MTW simulations using the dry weight at pupation as an endpoint showed a similar picture, although with higher EP_x values since the endpoint was less sensitive (supporting information,

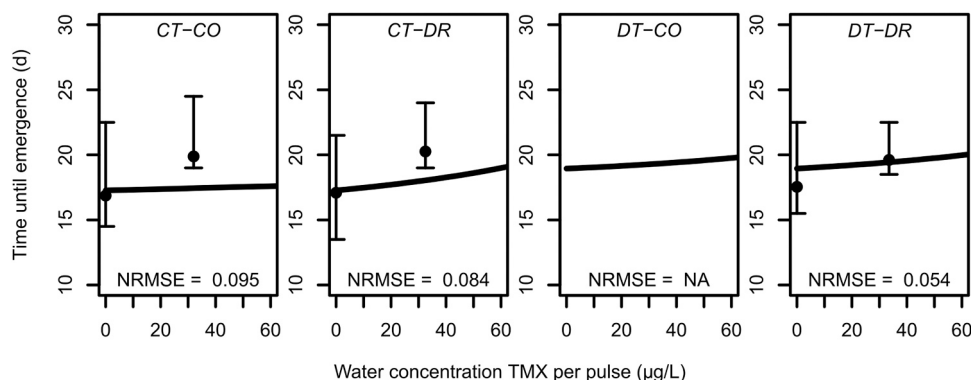


Fig. 4. Time-until-emergence data from the validation experiment and corresponding model simulations for an *in silico* recreation of the experimental conditions. Each column represents one of the four temperature-exposure pattern combinations applied in the validation experiments. Simulated data are presented as solid lines whereas measured data are presented as filled circles (mean) with whiskers (range). Indicated test concentrations correspond to analytically measured concentrations of thiamethoxam 1 h after spiking (mean of two pulses). Note that data were only available for the lower test concentrations (30 $\mu\text{g/L}$ nominal) since not enough individuals were able to successfully emerge at the higher concentrations (60 $\mu\text{g/L}$ nominal). No experimental data were available for scenario DT-CO as validity criteria were not met for this part of the test. Normalized root mean squared errors (NRMSE) are presented as a goodness-of-fit metric.

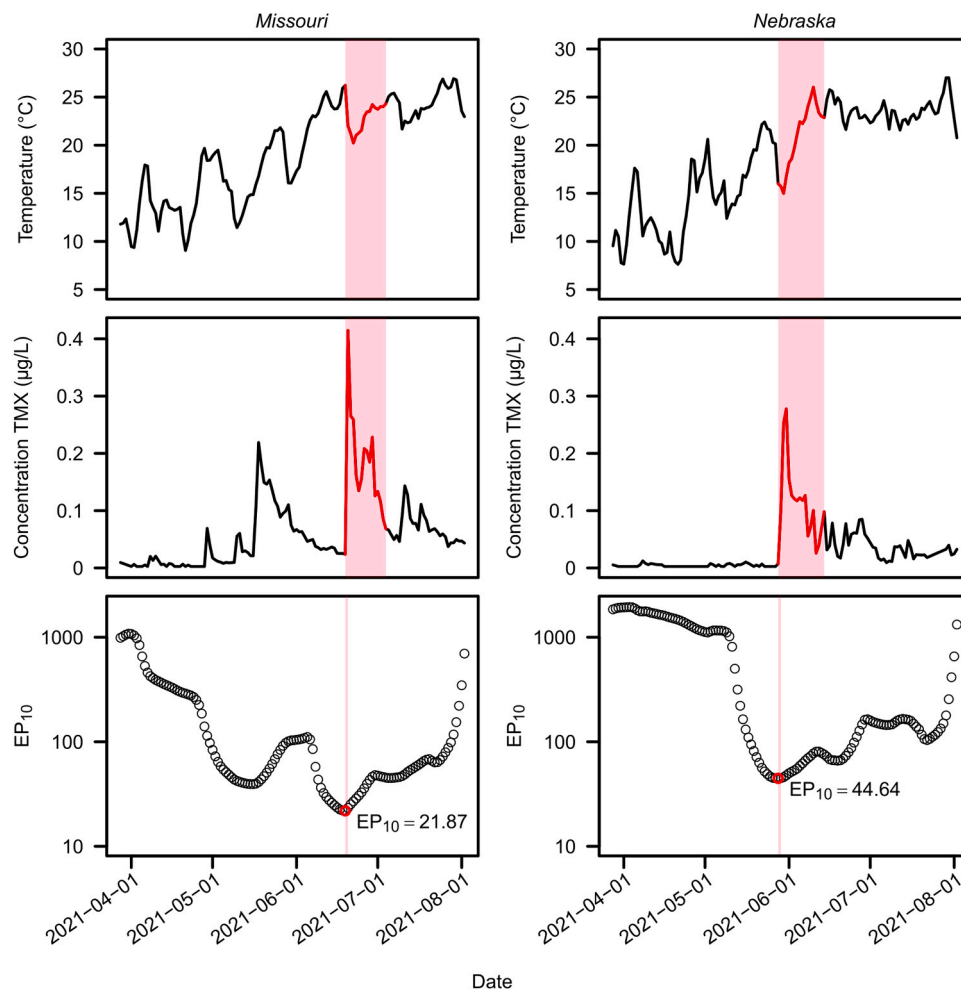


Fig. 5. Dynamic exposure scenarios for streams in Missouri and Nebraska with exposure multiplication factors leading to a 10 % effect (EP_{10}) in the survival at pupation, computed for each day as a potential starting day of the chironomid's life cycle, simulated with the temperature-dependent DEB-TKTD model. The minimum EP_{10} and the associated worst-case exposure window are highlighted in red for both scenarios.

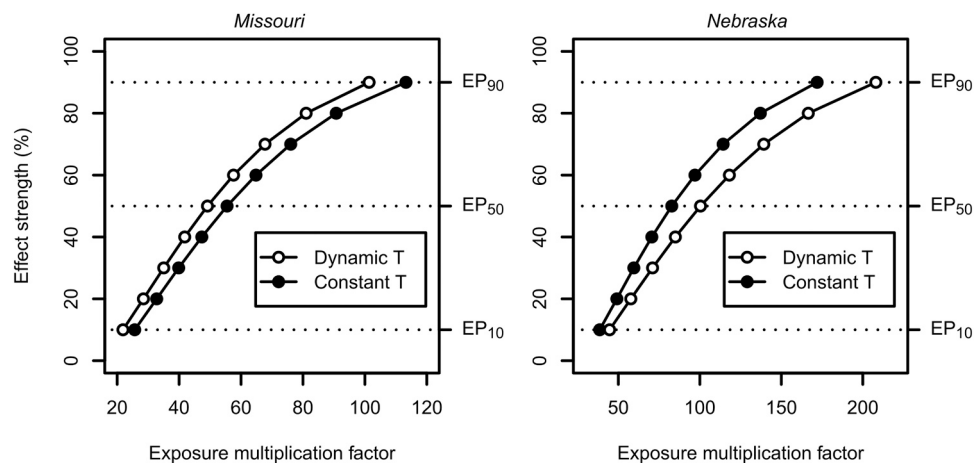


Fig. 6. Relative effect strengths (10–90 %) on the survival at pupation in function of exposure multiplication factors applied to the most adverse exposure window per scenario, as indicated by the lowest EP_{10} , using either the measured dynamic temperature profile, or a constant temperature of 20 °C. EP_x values (between EP_{10} and EP_{90}) can be derived from this figure by reading the exposure multiplication factor needed to achieve x % effect.

Figures S13, S14, and S15).

4. Discussion

4.1. Toxicity of thiamethoxam in *C. riparius*

A comparison of the observed effects of TMX to *C. riparius* in this study to those of an earlier study by [Saraiva et al. \(2017\)](#), using the same test substance and species in a similar test design (although without temperature variations), shows a high degree of agreement. Endpoints that were clearly affected in both studies are growth and survival. These effects could be reproduced successfully using our DEB-TKTD model with a pMoA on feeding. Notably, when reproducing the calibration experiments, the model also predicted a slight increase in the time until emergence with increasing concentration levels of TMX, an effect that was not apparent from the data (possibly because the effect slope was too subtle to be statistically detectable in our setup). Likewise, whereas [Saraiva et al. \(2017\)](#) observed a lower success rate for the emergence of chironomids with increasing TMX concentrations, no delay effects in the time until emergence were found among the individuals that emerged successfully. In our validation experiment, however, a clear increase in the development time was observed in the 30 µg/L pulse scenarios whereas – due to the high mortality – no data could be obtained for the 60 µg/L pulse scenarios. Here, the effect was even more pronounced in the experimental data versus the model simulations. The observed discrepancy between the calibration and the validation data for this endpoint suggests that the timing of exposure (pulses) plays an important mechanistic role, not perfectly explained by the model. Nevertheless, this indicates that developmental effects are (at least qualitatively) captured correctly by the model, providing some additional proof for the selected assumptions considering the trigger for pupation and the chosen pMoA. It is noteworthy, that while the pMoA on feeding was selected due to its superior data fit compared to other commonly used pMoAs, the same pMoA was also used in a study by [Gergs et al. \(2021\)](#) to describe developmental and survival effects of 3 functionally related insecticides, imidacloprid, thiacloprid, and flupyradifurone, in *C. riparius*, further supporting our decision on the pMoA choice.

4.2. Size-dependent survival sensitivity

In search of the most appropriate TKTD implementation to fit the experimental data, we used established DEB-TKTD formulae as specified by [Jager et al. \(2023\)](#) as a starting point. These include a one-compartment model for the scaled damage, a stochastic death model for survival, and a stress function linking the scaled damage to physiological processes to account for sublethal endpoints. While commonly used feedbacks between the physiological model and the scaled damage were initially assumed (i.e., a size correction factor for the dominant rate constant k_d accounting for changes in the surface-to-volume ratio, and a dilution-by-growth term), the timing of the onset of survival effects was captured best when removing the size-dependence term for k_d (supporting information, section S7, Figure S4). Despite this first improvement, the model still missed the survival effects observed at the lowest test concentration in three of the four temperature setups. With a closer look at the experimental data, the shift in the onset of survival effects at the lower test concentrations hints at a relatively slow damage build-up and, thus, a late exceedance of the threshold for survival z_s . However, considering that exposure concentrations steadily declined during the course of the experiment, it appears that a k_d suitable to capture the timing and magnitude of effect at higher concentrations leads to a damage kinetic that is too slow to ever exceed z_s at the lowest test concentrations. Hence some kind of increase in sensitivity to TMX during development was needed to also capture these low-concentration effects. Here, we linked sensitivity to the structural body length (as a proxy for developmental state) by adding a size-dependence term to z_s . This assumption led to a better fit to the

survival data, especially at low concentrations. While we lack a solid explanation for the observed increase in sensitivity during growth, we hypothesize that it is likely related to an increase of nAChRs – i.e., the target site where neonicotinoids act – during the chironomids' development. The idea that neonicotinoid sensitivity in chironomids may change throughout development due to changes in nAChR density and/or binding affinity was also brought forward by [Maloney et al. \(2021\)](#), although their study design, comparing nAChR binding affinities of imidacloprid in larvae versus adults (though not among different larval stages), was not comprehensive enough to validate this hypothesis because toxicity was not assessed concurrently. A study specifically investigating the receptor density and sensitivity of different aged larvae (e.g., with pulsed exposures at different ages) would help to further investigate this hypothesis. Finally, the fit was improved even further by removing the dilution-by-growth term from the TK model which according to [Jager et al. \(2023\)](#), is not always applicable when modeling the dynamics of damage (instead of modeling the internal concentrations explicitly). Nevertheless, the improved fit should not be seen as definitive evidence against dilution by growth in our study. It is possible that dilution by growth has taken place, but that its exclusion compensates for other unknown feedbacks. One such feedback could be that damage repair mechanisms can become saturated at high damage levels (see, e.g., [Klanjscek et al. \(2016\)](#)) leading to a deviation from the linear repair kinetics that follow from Eq. 3. Since a precise quantification of such processes is very difficult, however, we have instead used the exclusion of the dilution-by-growth term as the simplest possible option within the framework of the established formulae to match the data.

4.3. Temperature effects in *C. riparius*

Overall, the temperature-related effects on the life history of *C. riparius* observed in this study correspond well with those of an earlier study by [Péry and Garric \(2006\)](#), investigating temperature dependencies in the same species in a range of 15–26.7 °C. Both studies reported faster growth and a shorter time until emergence with increasing temperatures, whereas no effects on the reproduction and the survival were found in the respective temperature ranges. While [Péry and Garric \(2006\)](#) found a linear approximation fitting to describe how the growth rate relates to temperature, they acknowledged that the Arrhenius equation would also have been an appropriate choice, especially when considering a wider temperature range. In this study, we used an Arrhenius model with an upper temperature boundary, primarily since this submodel was already present in the existing AmP database entry for *C. riparius* which our model was structurally based on. The model delivered a good approximation for the temperature dependencies observed in the control data.

4.4. Temperature dependence in TKTD parameters

In previous studies, temperature dependencies in TKTD parameters have been addressed in various ways. While there clearly is a need to account for temperature in the TKTD processes, it is still an open question which parameters should be temperature corrected and which can be considered temperature independent ([Jager and Ashauer, 2022](#)). Two recent studies modeling temperature-dependent survival toxicity using the same stochastic death assumption used in our study, though without consideration of sublethal effects and physiological feedbacks to the damage dynamics, both found that a temperature dependence of both the toxicokinetics (via the dominant rate constant, here denoted k_d) and the toxicodynamics (via the lethal effect strength parameter, here denoted b_s) described their results best ([Mangold-Döring et al., 2022](#); [Rakel et al., 2022](#)). However, whereas [Rakel et al. \(2022\)](#) demonstrated that a physiologically derived Arrhenius temperature, applied to k_d and b_s in the same way, was suitable to model the survival of *Cloeon dipterum* exposed to imidacloprid, [Mangold-Döring et al. \(2022\)](#) showed that Arrhenius temperatures calibrated independently for k_d and b_s delivered

the best results in their study, assessing the effects of flupyradifurone and imidacloprid in *Gammarus pulex*. Another study by Heugens et al. (2003), in which k_d and b_s were fitted independently at multiple temperatures, also suggested that a temperature-dependence in both parameters is likely needed.

From a DEB-theoretical point of view, temperature corrections should be applied to all rates, i.e., time-based processes such as physiological rates in the core DEB model (Kooijman, 2009). While the parameter b_s is expressed in a unit *per day*, it is different from other rate parameters in that it represents a (time-based) probability rather than an energetic flux or chemical transport process, which raises the question if the same type of temperature correction should be applied. Furthermore, if a temperature correction for b_s is applied in addition to k_d , this also raises the question if and where a temperature correction should also be applied in the sublethal effect parameters in a DEB-TKTD model, none of which carry a unit *per day*.

With these ambiguities unresolved, we decided to follow a more data-driven approach, systematically testing which of the above-described assumptions best fit our experimental findings (see supporting information, section S6), ultimately resulting in the exclusion of a temperature correction of b_s , and the assumption of an independent Arrhenius temperature for k_d . Note that this result should not be seen as proof for either of the assumptions since most combinations of assumptions tested led to acceptable fits as long as a temperature-dependence of k_d was present. We believe our study contributes substantially to the understanding of how toxicity is modulated by temperature. Nonetheless, a more comprehensive theoretical and empirical approach, including multiple chemical compounds and species, could help to further elucidate this subject.

4.5. Simulations of realistic exposure scenarios

We determined EP_x values for both exposure scenarios using either a constant temperature (20 °C), which is currently the standard approach as suggested by Sherborne et al. (2020), or the actual measured temperature profile for each scenario. The latter approach was chosen to take full advantage of the newly parametrized temperature-dependence in the DEB-TKTD model, moving one step further from a lab mimic scenario toward a realistic field situation, thereby reducing some of the uncertainty associated with extrapolating effects from lab to field conditions. Comparing the results, this new approach led to a 15 % lower EP_{10} for the Missouri scenario and a 16 % higher EP_{10} for the Nebraska scenario when compared to the fixed temperature approach. The direction of the offset can be explained by the fact that the model predicts stronger effects of TMX at higher temperatures, and the measured temperatures during the strongest exposure pulse in the Missouri scenario were above 20 °C whereas the opposite was the case in the Nebraska scenario. Overall, our results illustrate that while adding more realism to the MTW method, this alternative approach does not systematically result in either a more conservative or a less conservative outcome for EP_x , a metric to be potentially used in a risk assessment. Importantly, however, the added realism reduces the uncertainty associated with effect extrapolation from constant (laboratory) conditions to the dynamic environment in the field, thereby providing an important refinement to the risk assessment. More broadly, this demonstrates how mechanistic effect models, which by their design can be used flexibly to simulate both stable laboratory conditions as well as more realistic field situations, can be used as needed to incorporate specific environmental factors deemed relevant to improving risk assessment.

5. Conclusions

We demonstrated, how a DEB-based TKTD model describing the effects of TMX in *C. riparius* could be successfully parametrized with data from literature (primarily informing the unstressed physiological model) and chronic life-history experiments with the chemical. Additionally,

temperature-dependent effects could be easily accounted for in the model using Arrhenius-based temperature functions both in the physiological model and the TKTD part. Overall, we consider the model suitable to extrapolate effects to untested dynamic conditions of TMX exposure and temperature. However, we acknowledge that the model may still be further improved as new insights into DEB modeling of holometabolous insects and temperature dependence in TKTD modeling emerge. Finally, we showed how the MTW approach, used to evaluate realistic exposure patterns for risk assessment, can be easily adapted to account for dynamic temperature profiles, adding realism to the approach, and thereby reducing the uncertainty when extrapolating effects from laboratory conditions to the field.

CRedit authorship contribution statement

Maxime Vaugeois: Writing – review & editing, Supervision, Resources, Project administration, Methodology, Investigation, Funding acquisition, Conceptualization. **Tido Strauss:** Writing – review & editing, Supervision, Software, Resources, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Conceptualization. **Nika Galic:** Writing – review & editing, Supervision, Resources, Project administration, Methodology, Investigation, Funding acquisition, Conceptualization. **Silke Classen:** Writing – review & editing, Validation, Supervision, Resources, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization. **Josef Koch:** Writing – original draft, Visualization, Validation, Supervision, Software, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization. **Natalie Dallmann:** Writing – review & editing, Visualization, Software, Methodology, Investigation, Formal analysis, Data curation. **Daniel Gerth:** Writing – review & editing, Validation, Methodology, Investigation, Formal analysis, Data curation.

Declaration of Generative AI and AI-assisted technologies in the writing process

During the preparation of this work the authors used the DeepL translation tool in order to improve sentence structure. After using this tool, the authors reviewed and edited the content as needed and take full responsibility for the content of the publication.

Declaration of Competing Interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: Josef Koch reports financial support and article publishing charges were provided by Syngenta. Silke Classen reports financial support was provided by Syngenta. Daniel Gerth reports financial support was provided by Syngenta. Natalie Dallmann reports financial support was provided by Syngenta. Tido Strauss reports financial support was provided by Syngenta. If there are other authors, they declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data Availability

Data will be made available on request.

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Appendix A. Supporting information

Supplementary data associated with this article can be found in the online version at [doi:10.1016/j.ecoenv.2024.116355](https://doi.org/10.1016/j.ecoenv.2024.116355).

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